

# Fundamental Concepts of Computer Graphics

Lecture 01

W.A.G.Nilnuwani,

Department of Information Technology,

Faculty of Information Technology,

University of Moratuwa

*IN2601 Computer Graphics*

# Outline

- Introduction
- Major Areas in Computer Graphics(CG)
- Historical Milestones
- Application Areas in CG
- Computer Graphics Vs Computer vision
- Raster and Vector Graphics
- Graphics pipeline

# Introduction to Computer Graphics

- Computer graphics describes any use of computers to create and manipulate images.
- The main result of computer graphics is a **picture or visual image**.
- Graphics can be:
  - **2D (two-dimensional)** - flat images like drawings or photos
  - **3D (three-dimensional)** - images with depth like models or animations
- Images can be:
  - **Created from scratch (synthetic images)**
  - **Edited or improved from real photographs**

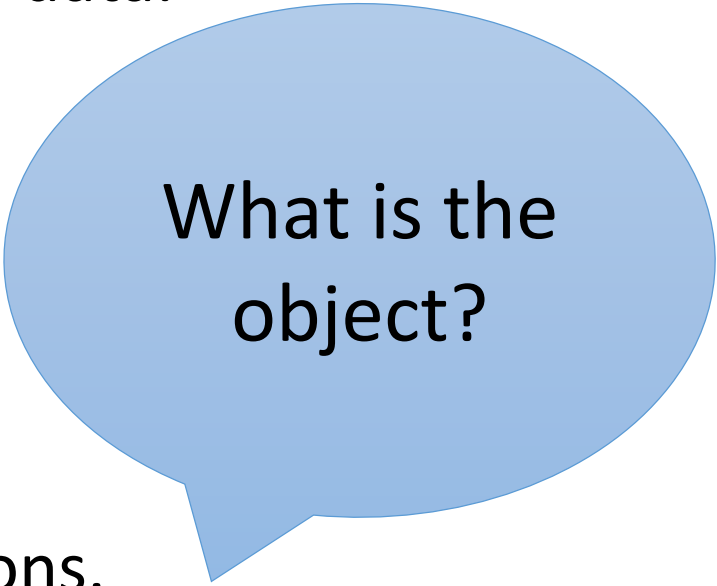
# Major Areas in Computer Graphics

## 1. Modelling

- Modelling is the process of mathematically describing the shape and structure of objects in a computer using geometric data.
- Defines the object's geometry.
- Uses coordinates (2D:  $x, y$  | 3D:  $x, y, z$ ).
- Includes vertices, edges, surfaces, curves, etc.

Example:

- Creating a 3D cube using 8 vertices and 6 faces.
- Drawing a vector circle using mathematical equations.

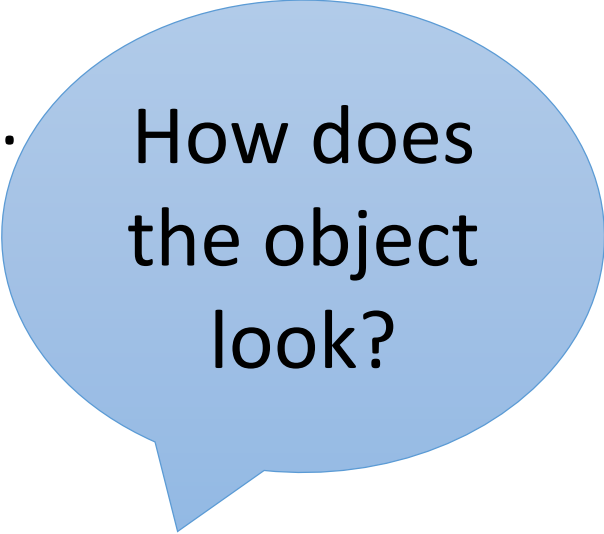


What is the object?

# Major Areas in Computer Graphics

## 2. Rendering

- Rendering is the process of generating a final visual image from the modelled data so that it can be displayed on a screen.
- Converts stored geometric data into pixels.
- Applies lighting, shading, color, texture.
- Produces a 2D image as output (even for 3D scenes).
- Example:
- Displaying a shaded 3D object on screen.
- Drawing a 2D image with proper colors and effects



How does the object look?

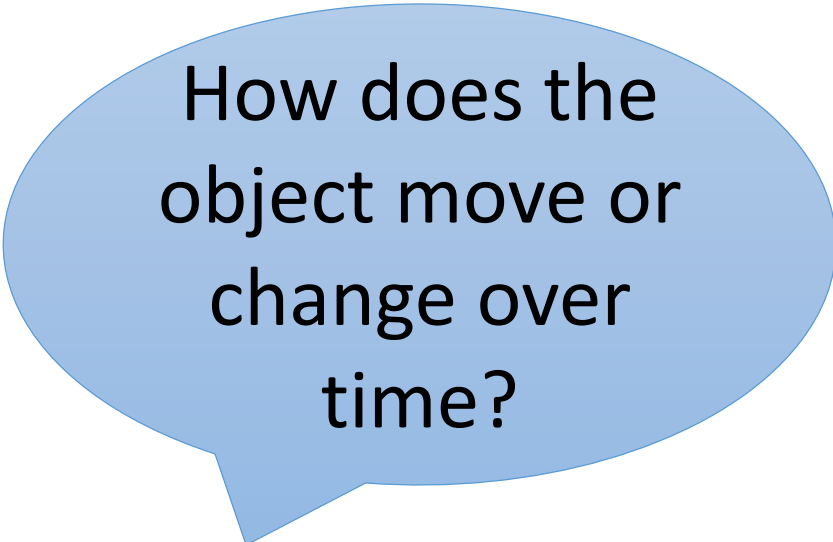
# Major Areas in Computer Graphics

## 3. Animation

- Animation is the process of creating the illusion of motion by changing graphical objects over time.
- Updates object position, rotation, scale, shape, or properties.
- Displays frames sequentially.
- Can be 2D or 3D.

Example:

- Moving a car across the screen.
- A character walking in a game.

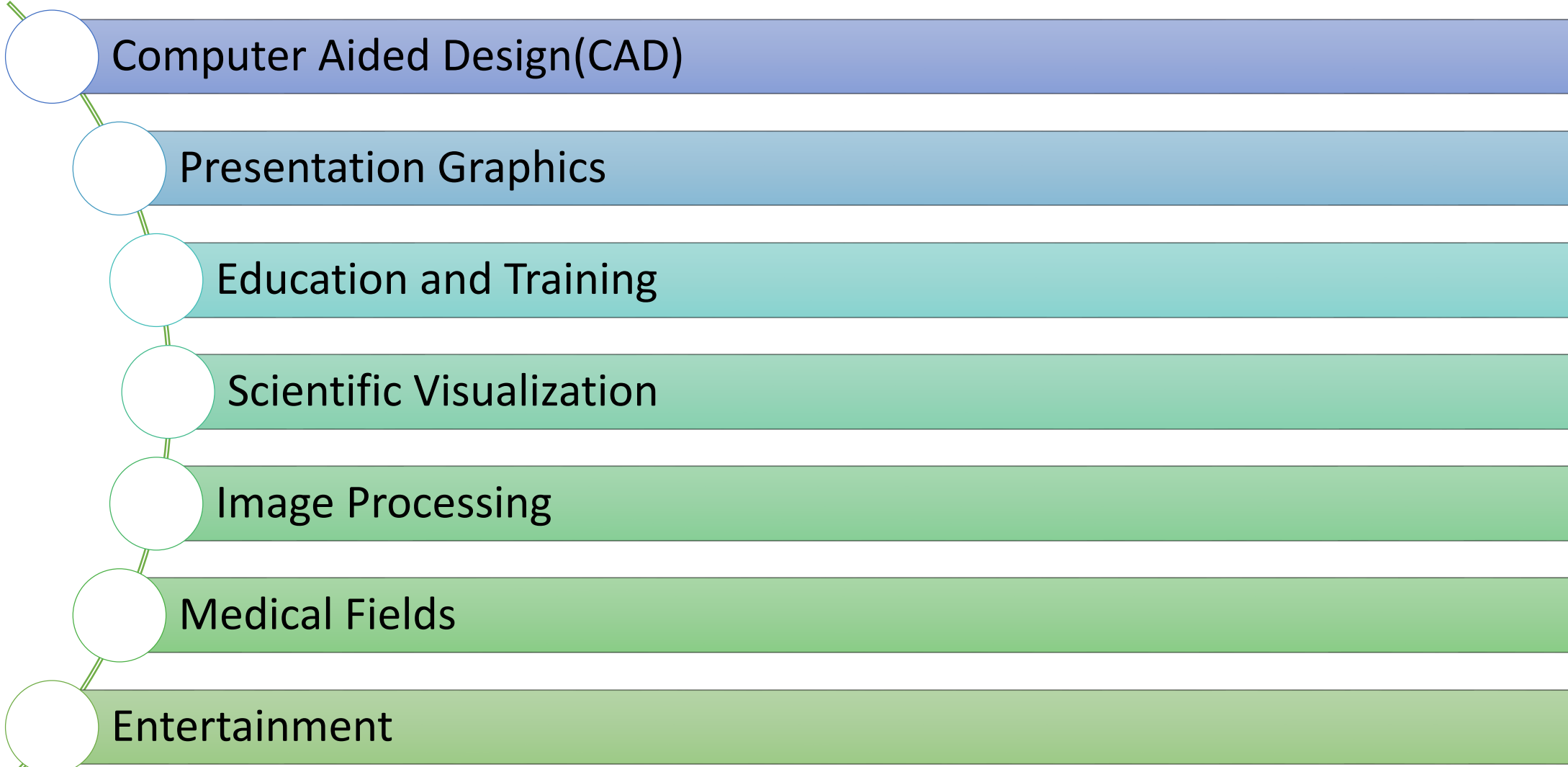


How does the object move or change over time?

# Activity

1. Is Photoshop computer graphics?
  2. Is PUBG computer graphics?
  3. Is Google Maps computer graphics?
- Justify why.

# Application Areas



Computer Aided Design(CAD)

Presentation Graphics

Education and Training

Scientific Visualization

Image Processing

Medical Fields

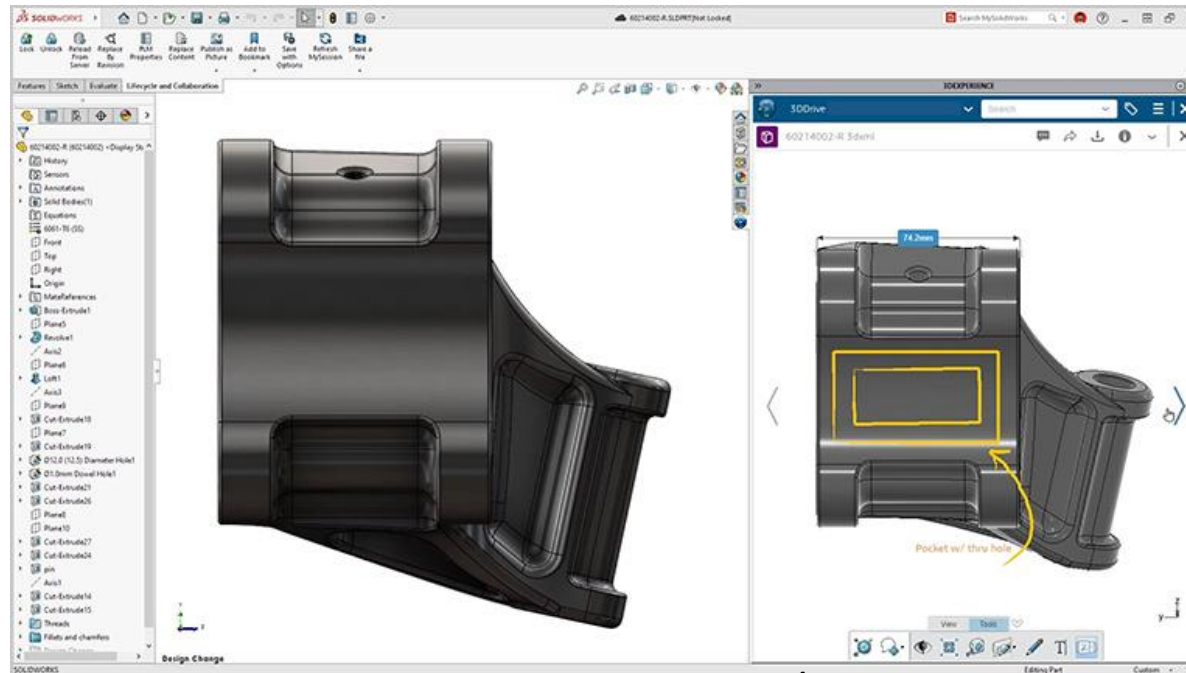
Entertainment



# CAD/CAM

## 1. CAD (Computer-Aided Design)

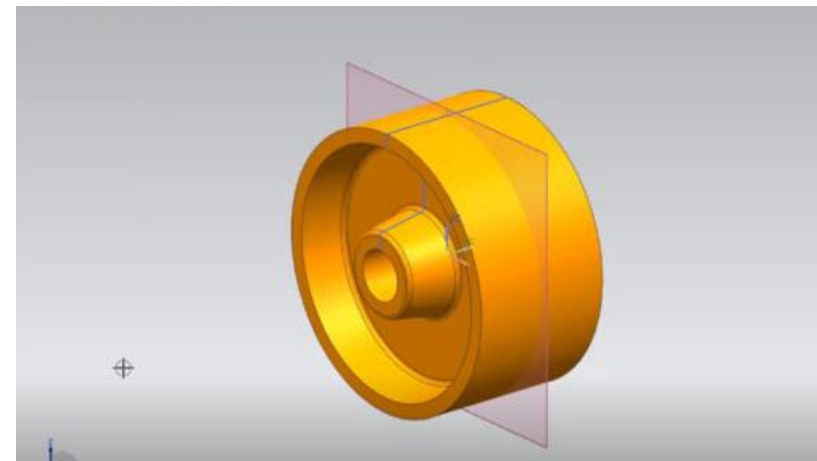
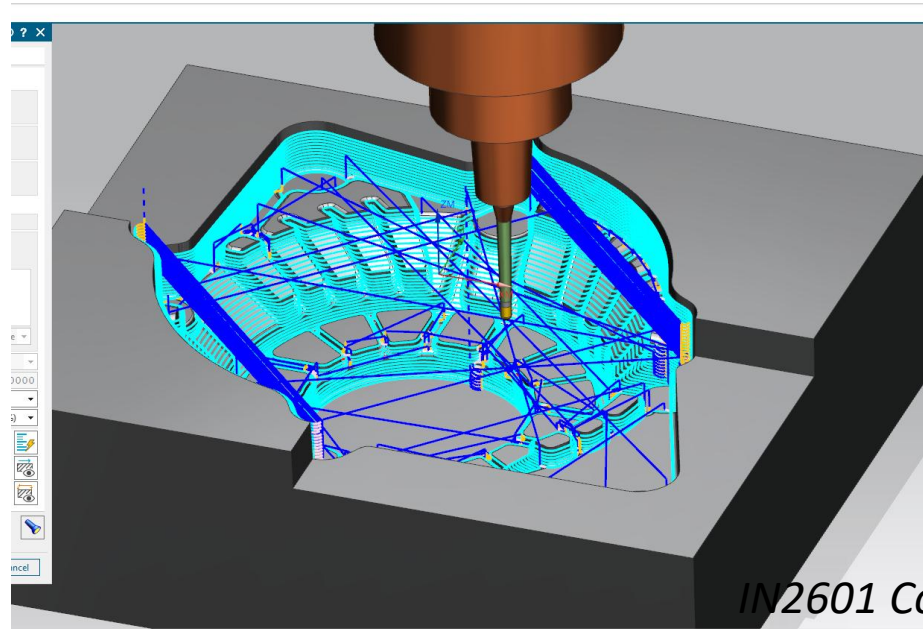
- Used to create **2D drawings** and **3D models** of real-world products.
- Common in **engineering and architecture**.



# CAD/CAM

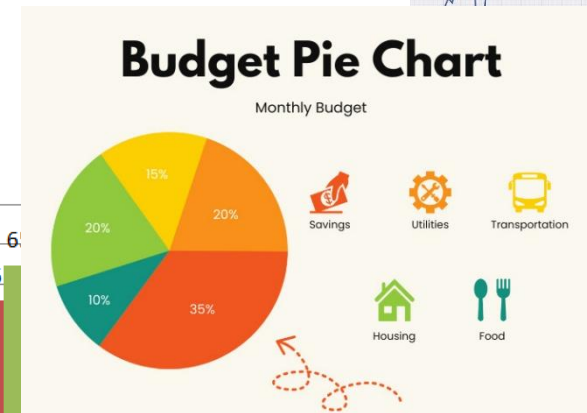
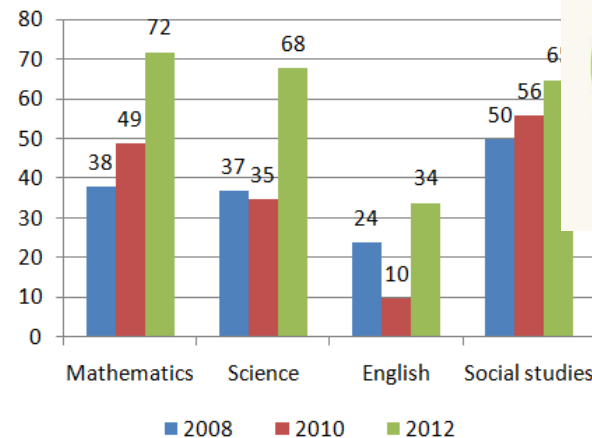
## 2. CAM (Computer-Aided Manufacturing)

- Uses computer designs to **control machines and manufacture products**.
- Example: Designing a house plan or a car engine using software.



# Presentation Graphics

- Used to **present data in a visual way.**
- Helps to understand:
  - Mathematical data
  - Financial data
  - Scientific data
  - Economic data
- Common examples:
  - **Bar Charts**
  - **Pie Charts**
  - **Line Graphs**
- Purpose: Makes complex data **easy to understand and compare.**



# Education and Training

- Computers can create **models of real-world systems** (physical, financial, or economic systems).
- These models are used for **learning and training**.
- Special programs called **simulators** help people practice safely.
- Example:
  - Driving Simulator
  - Flight Simulator
  - Medical Simulator

# Education and Training

- Example:
  - Driving Simulator





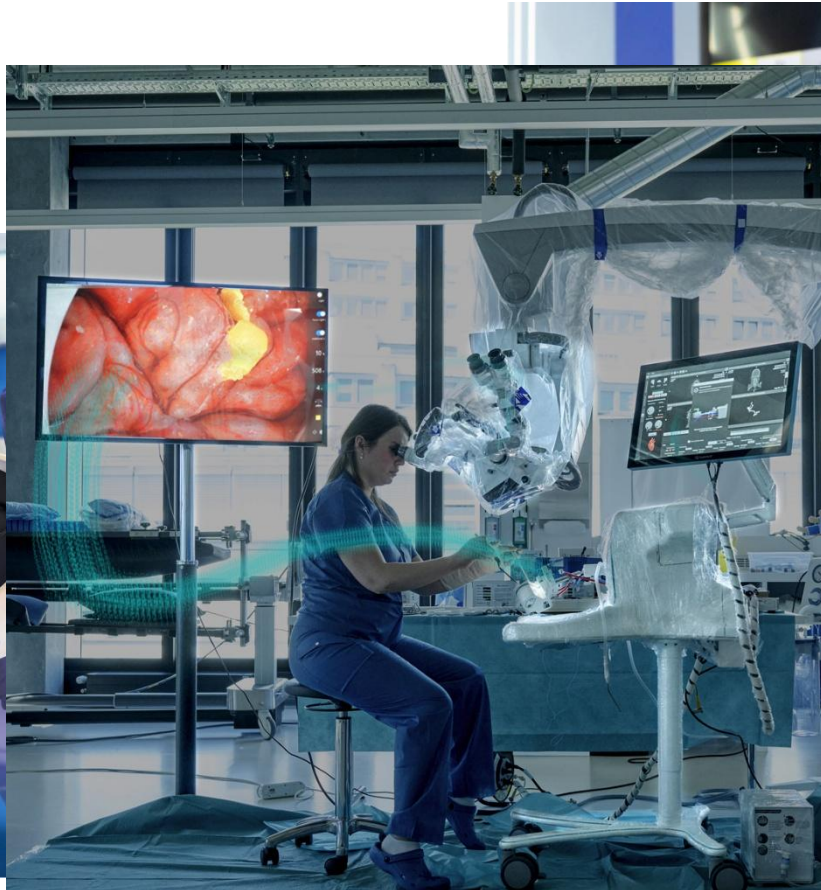
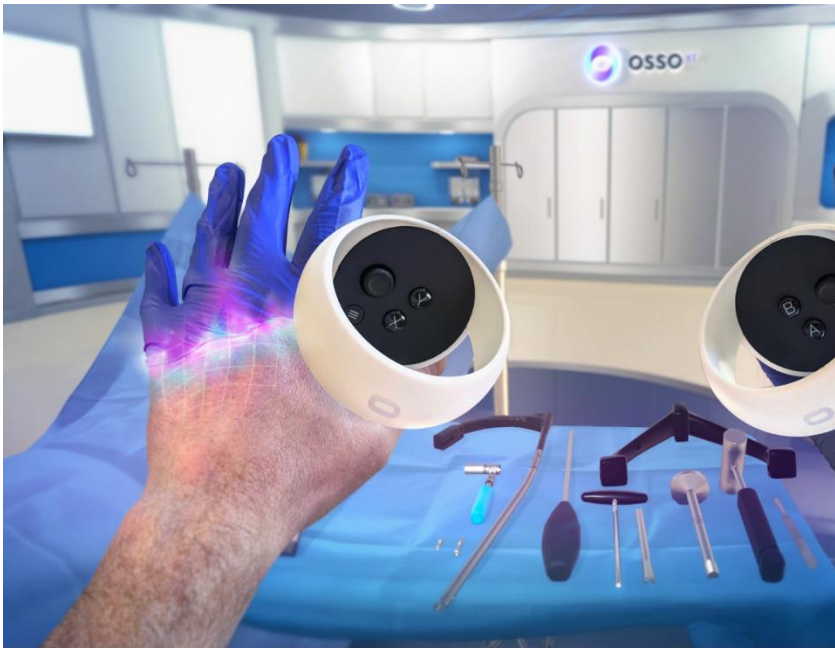
# Education and Training

- Example:
  - Flight Simulator



# Education and Training

- Example:
  - Medical Simulator



IN2601 Computer Graphics

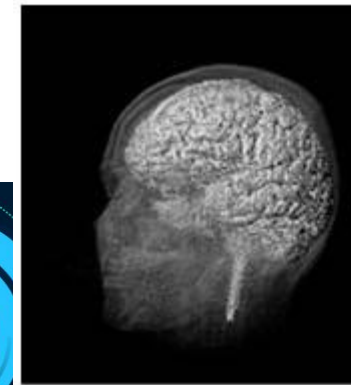
# Scientific Visualization

- Uses computer graphics to **show scientific data visually**.
- Helps to understand **large and complex data** easily.
- Used in many fields like:
  - Medicine
  - Weather forecasting
  - Space research
  - Engineering



# Scientific Visualization

- Medicine



(a) Brain Opacity: 1.00  
Skin Opacity: 0.05



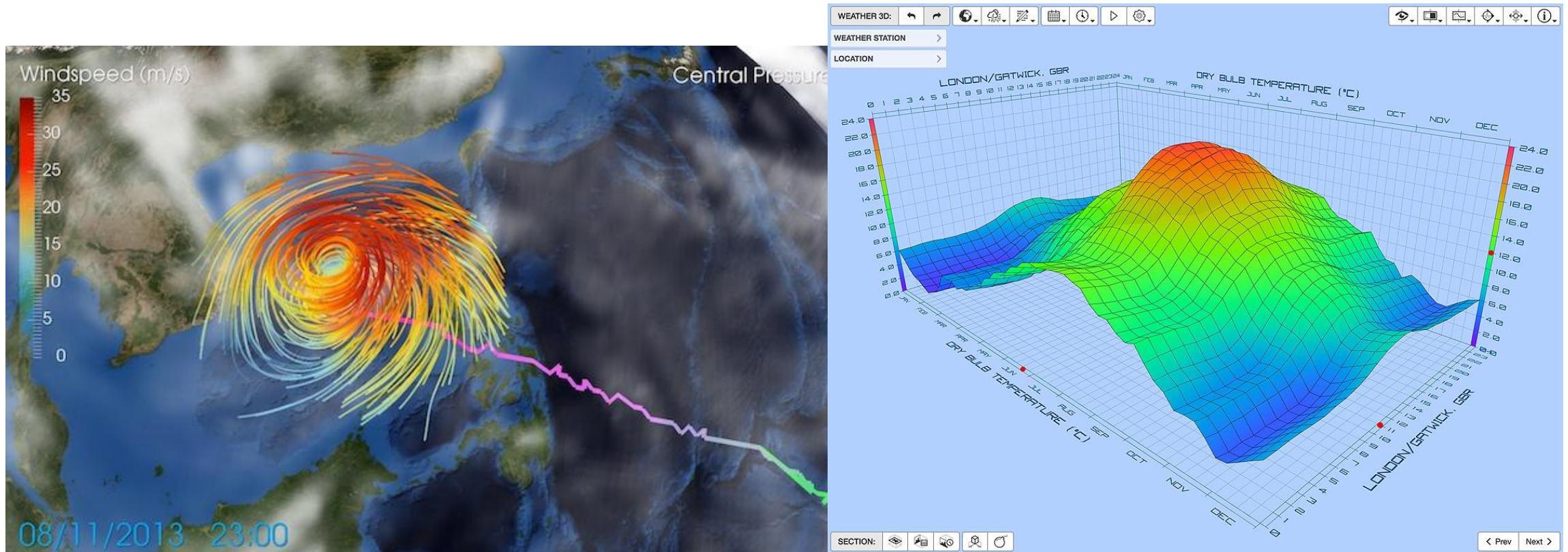
(b) Brain Opacity: 1.00  
Skin Opacity: 0.15



(c) Brain Opacity: 1.00  
Skin Opacity: 0.15

# Scientific Visualization

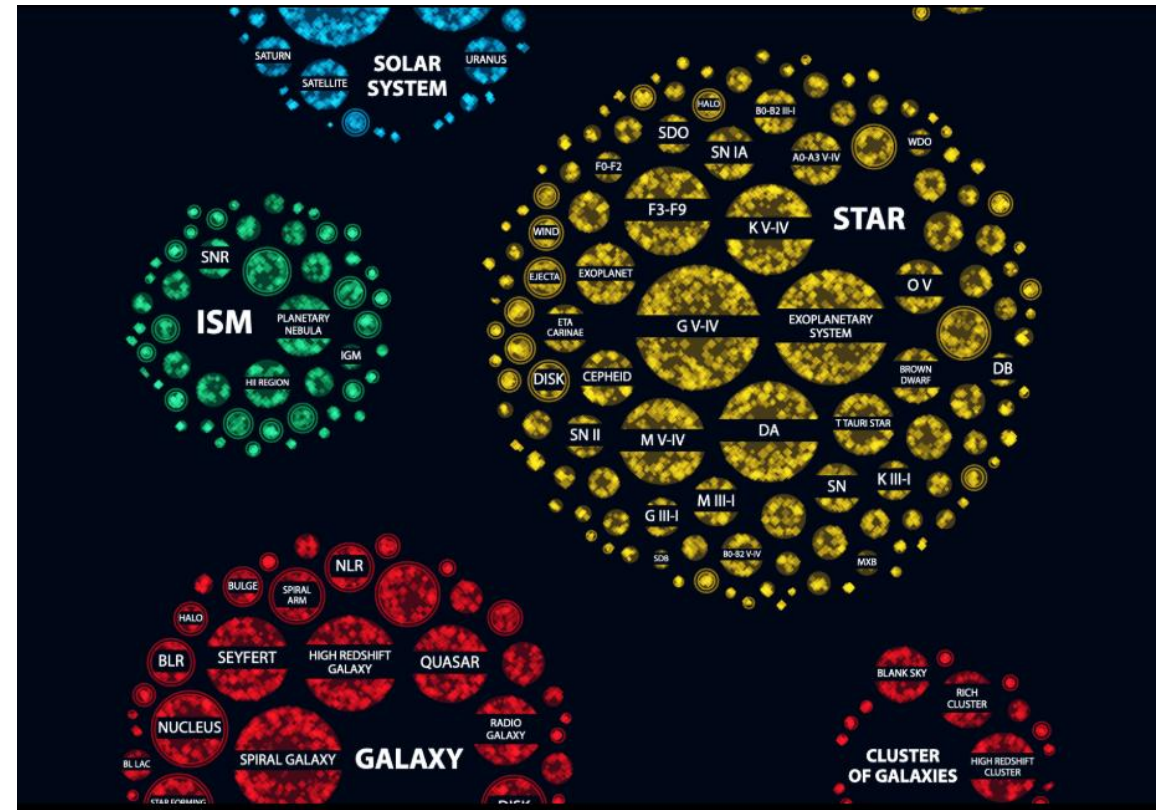
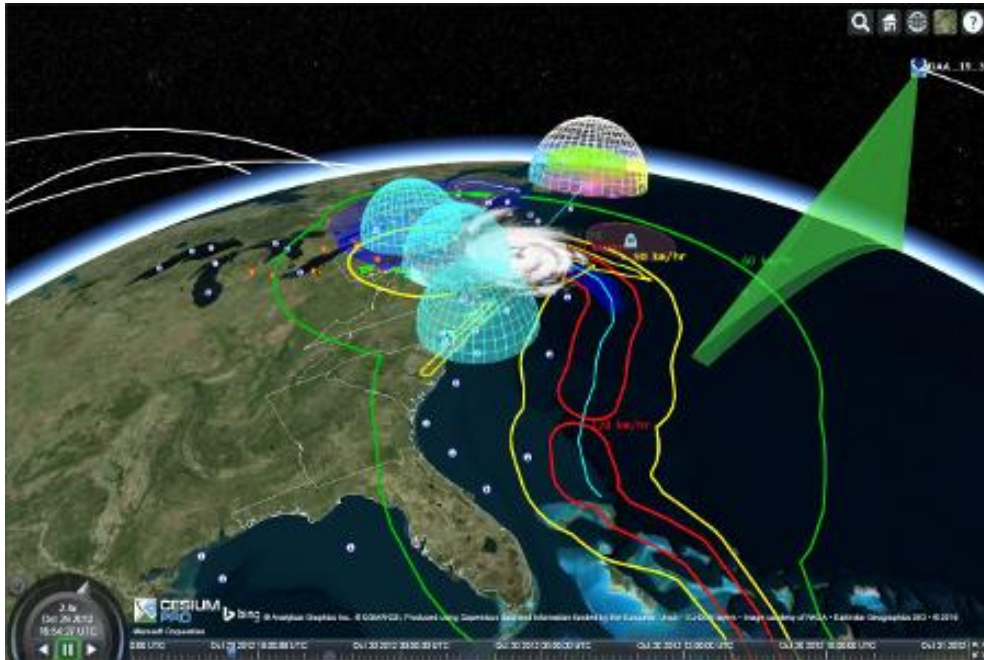
- Weather & Climate Visualization





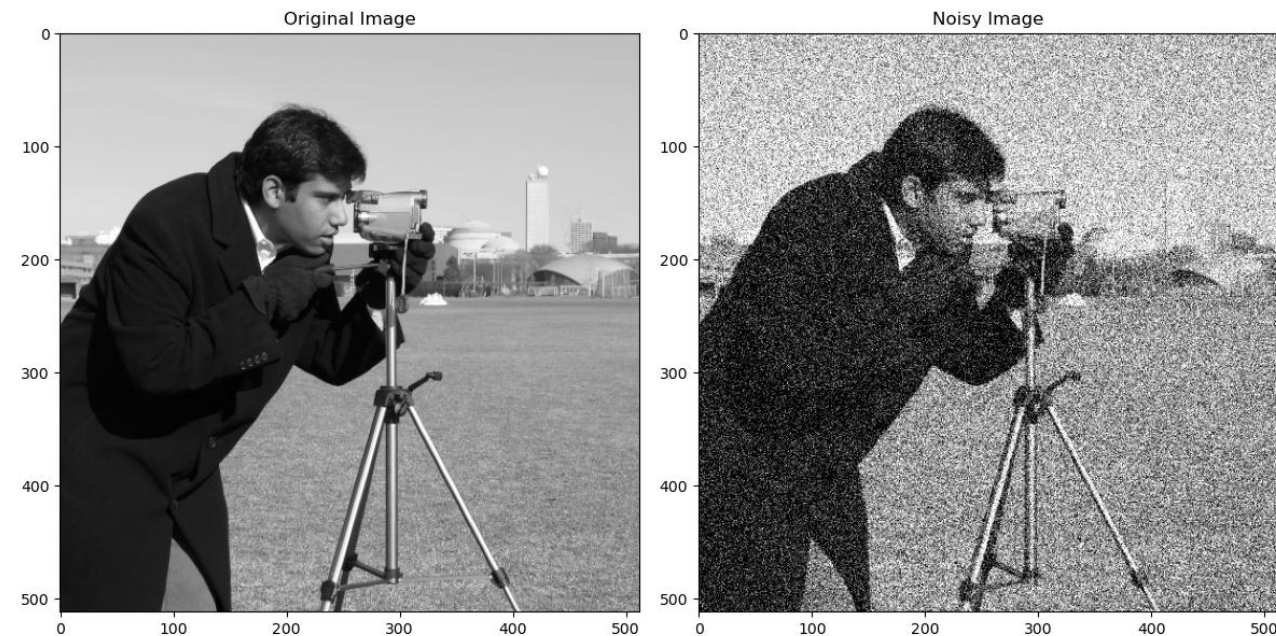
# Scientific Visualization

- Space & Astronomy Visualization



# Image Processing

- **Computer Graphics** creates new images while **Image Processing** edits or analyzes existing images.
- It improves or changes pictures using computer techniques.



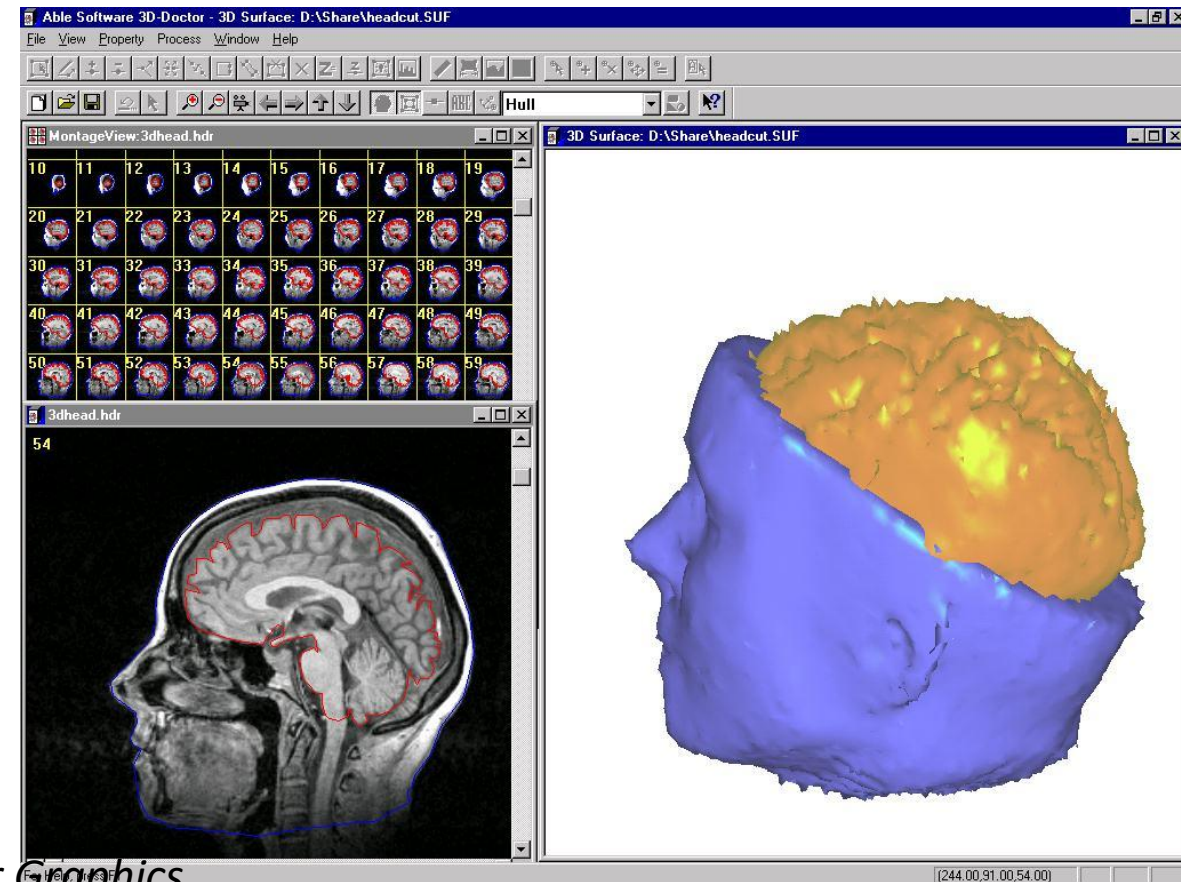
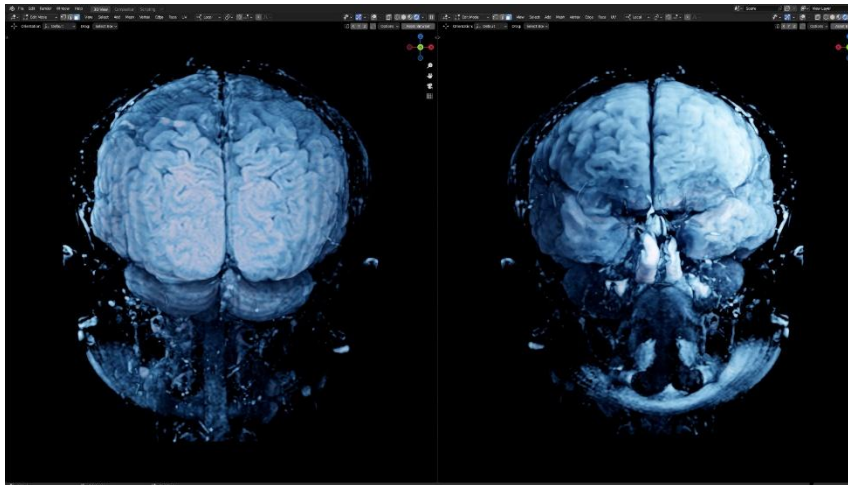
# Activity

- What is the difference between **creating an image** and **improving an existing image**?
- Provide a real-world example of each



# Medical Fields

- Computer graphics is used to **show the inside of the human body**.
- Helps doctors to:
  - View internal organs
  - Study body structures
  - Understand medical processes
  - Makes diagnosis and treatment easier.





# Activity

Explain **Medical Imaging** vs **Medical Simulation** using real world examples

# Entertainment

- Computer graphics is widely used in the entertainment industry.
  - Video Games
  - Cartoons
  - Animated Films

# Entertainment

## Video Games

- Creates realistic characters and environments
- Adds special effects and animations
- Example: 3D adventure and action games



# Entertainment

## Cartoons

- Uses 2D or 3D graphics
- Brings drawings to life through animation

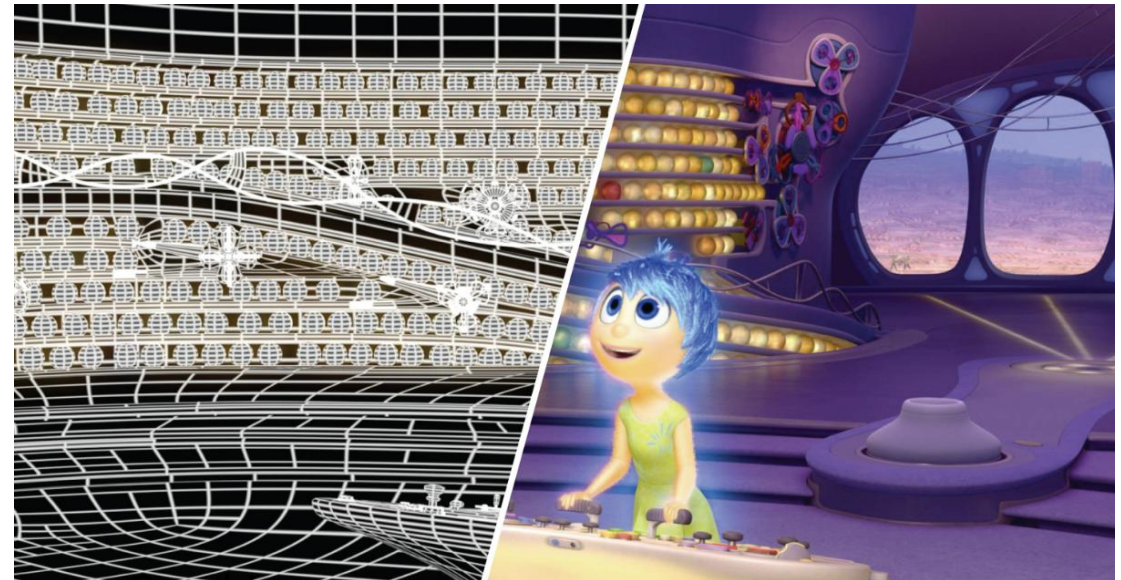




# Entertainment

## Animated Films

- Uses advanced 3D modelling and rendering
- Combines graphics, lighting, and sound
- Example: Full-length 3D animated movies



# Activity

Why do video games need **real-time rendering**, but animated films do not?

# Historical Milestones

- 1950s - Beginning
  - First images created using an oscilloscope, Simple waveform graphics
- 1960s - Research Stage
  - Used mainly in research & military, Sketchpad by Ivan Sutherland, Foundation for CAD
- 1970s - Graphics Pipeline
  - Traditional graphics pipeline developed, Growth from military and automotive industries
- 1980s - Core Algorithms
  - Major rendering algorithms developed, 1985 Rendering Equation by James Kajiya, Movie industry improved visual quality
- 1990s - Advanced Techniques
  - Realistic rendering and animation, 1995 Photon Mapping by Henrik Wann Jensen
- 2000s - Modern Graphics Era
  - Photorealistic graphics on normal computers, Powerful GPUs (driven by video games), Graphics + Computer Vision integration

# Activity

**Why did graphics improve after 1980s?**

# Activity

## **Why did graphics improve after 1980s?**

- GPU development
- Game industry funding
- Mathematical algorithms



# Activity

**Why did the video game industry push GPU development?**

# Activity

## **Why did the video game industry push GPU development?**

Video games require real-time rendering. To display smooth motion and high-quality graphics instantly, powerful GPUs were needed. As GPUs improved, other fields like movies, medical imaging, and simulation also benefited.

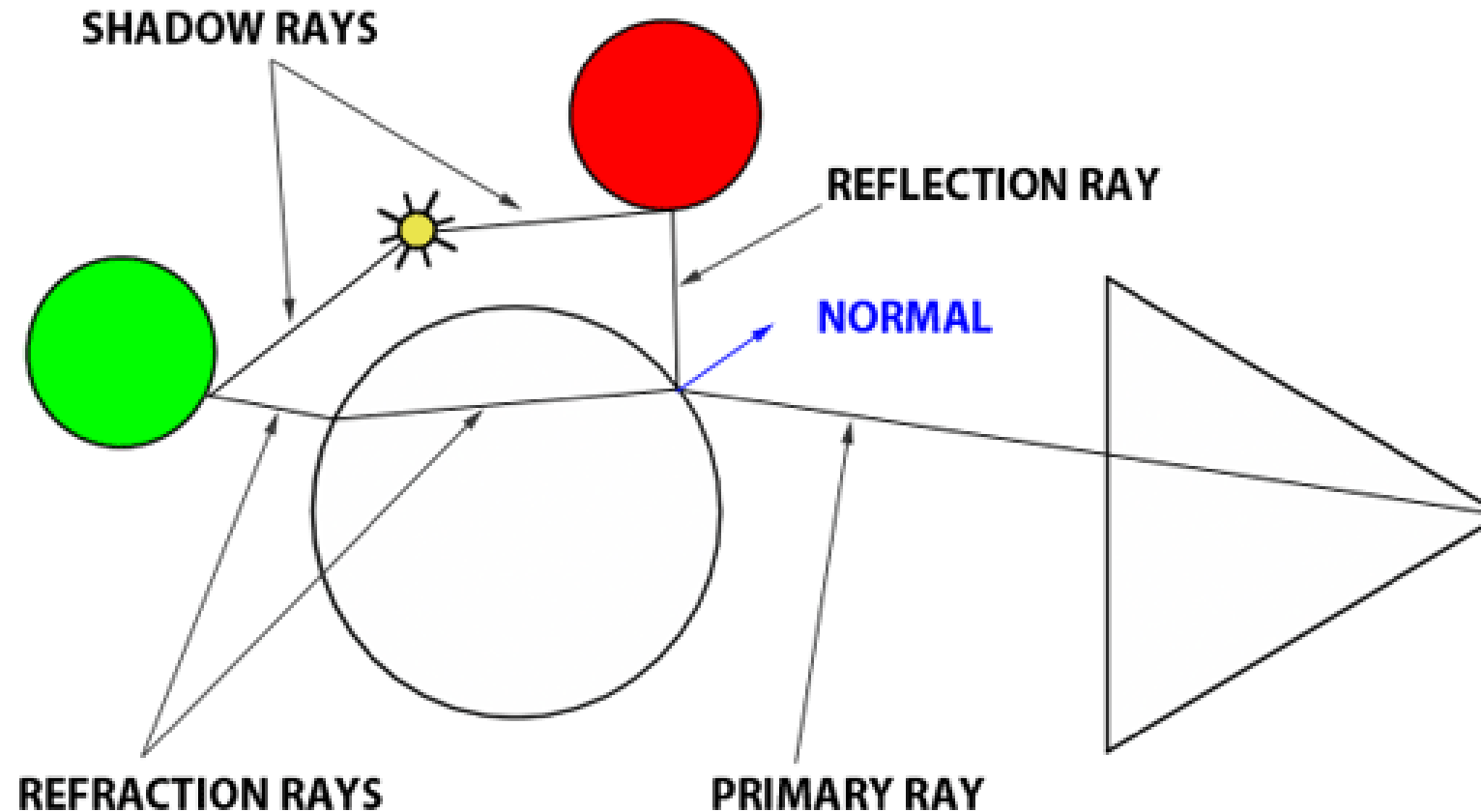
# Photoreal Rendering

- Photorealistic rendering is the process of generating images that look as realistic as real-world photographs. The goal is to make the image look like a **real photograph**.
- It follows the **science of light (optics)** to produce correct lighting and shadows.
- Uses special algorithms to simulate how **light travels**.
- Most modern methods are based on **Ray tracing**.

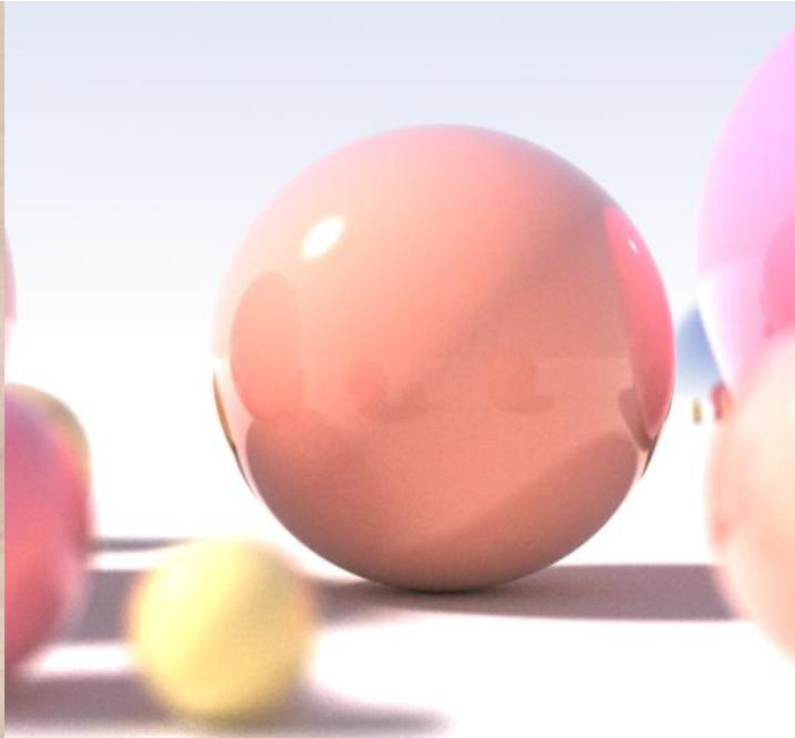
Ray tracing:

- Traces light rays from the camera (eye)
- Calculates reflections, shadows, and lighting
- Produces realistic images

# Photoreal Rendering – Ray Tracing



# Photoreal Rendering





# Activity

**Why is ray tracing slower than normal rendering?**

# Activity

## Why is ray tracing slower than normal rendering?

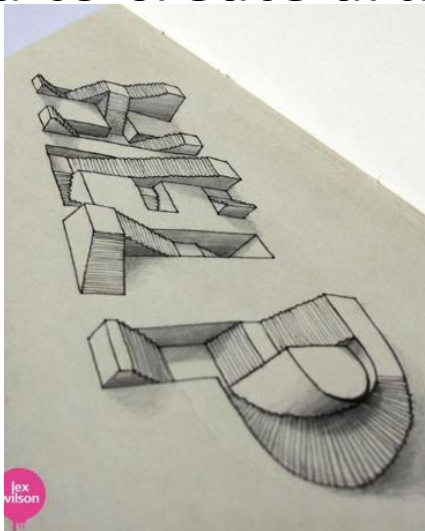
- Ray tracing is slower because it tries to copy how **real light works**.
- It follows many light rays from the camera to objects and checks:
  - Reflections (like mirrors)
  - Refractions (like glass)
  - Shadows
  - Light bouncing between objects
- All these steps need many calculations, so it takes more time and computer power.
- Normal rendering methods are faster because they use **simpler lighting tricks** instead of calculating real light behavior.

# Non-Photoreal Rendering (NPR)

- Rendering images in **artistic or non-realistic styles**.
- The goal is **not to look like a real photo**.
  - Artistic Styles
  - Communication Purpose
- **Abstraction** - Remove unimportant details
- **Ambiguity** - Hide some details intentionally
- **Emphasis** - Highlight important parts

# Non-Photoreal Rendering – Artistic Style

- Watercolor style
- Pencil sketch style
- Paint brush strokes
- Cartoon effects
- Used to create artistic and creative visuals.



# Activity

**Why would we remove details instead of adding more realism?**



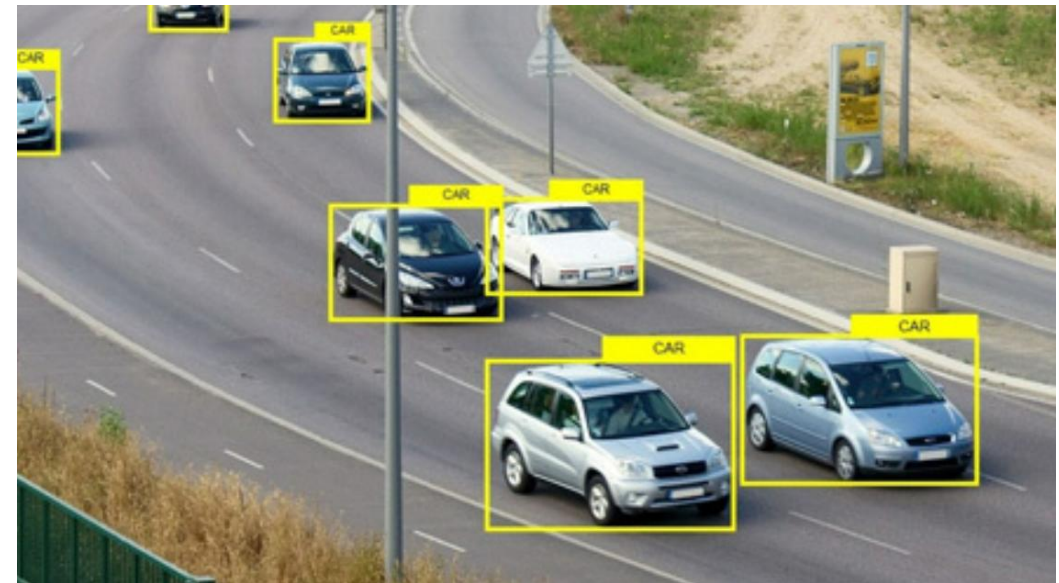
# Activity

## **Why would we remove details instead of adding more realism?**

- We remove details when they are not important.
- Too much detail can confuse people and make the picture harder to understand.
- By removing extra details, we can,
  - Make important parts easy to see
  - Help people focus on the main idea
  - Make the image clearer
- For example, in medical diagrams, small textures and extra colors are removed so doctors can clearly see the important body parts.

# Computer vision

- A field that helps computers **understand images and videos**.<sup>3</sup>
- It allows computers to **recognize objects, faces, shapes, and scenes**.
- From 2D Images to 3D Understanding.
  - Takes 2D images (photos or video frames)
  - Analyzes them
  - Understands the 3D real-world scene
- Example:
  - Self-driving cars detecting pedestrians
  - Face recognition systems
  - 3D reconstruction from images
- Vice versa of Computer Graphics



# Raster Graphics

- Raster graphics are made using a **grid of tiny squares called pixels**.
- Each pixel has:
  - A **color value**
  - A **position** on the screen
- Also called **Bitmap images**.

# Raster Graphics

What Affects Image Quality?

- Quality is determined by the total number of pixels(Resolution) and the amount of information in each pixel(colordepth).
- **Resolution** - Total number of pixels  
(More pixels = clearer image)
- **Color Depth** - Amount of color information in each pixel  
(More bits = more colors)

# Raster Graphics

- **Framebuffer - memory area** in the computer which stores pixel information (colors and brightness)
- **Color Systems**
  1. **Black & White System**
    - Uses **1 bit per pixel**
    - Pixel is either black or white
  2. **True Color System**
    - Uses **24 bits per pixel**
    - Can display millions of colors

# Raster Graphics

- Most Common file extensions of raster graphics are .jpg, .png,





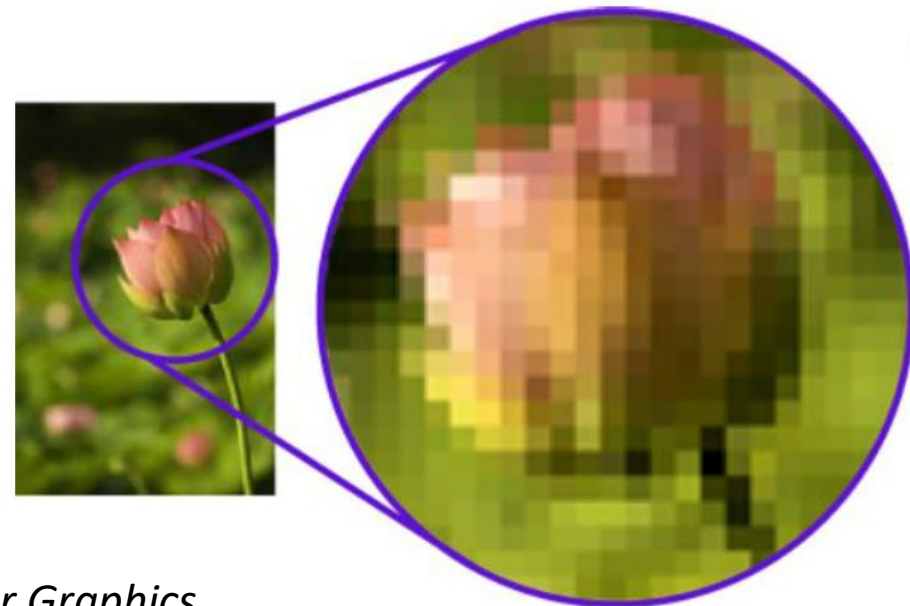
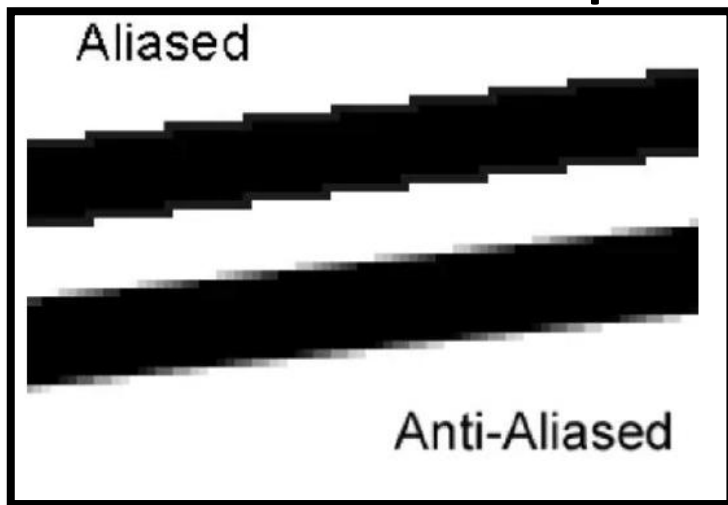
# Raster Graphics

- **What Happens When We Enlarge a Raster Image?**

- Raster images are made of **small square pixels**.  
When we zoom in or enlarge the image

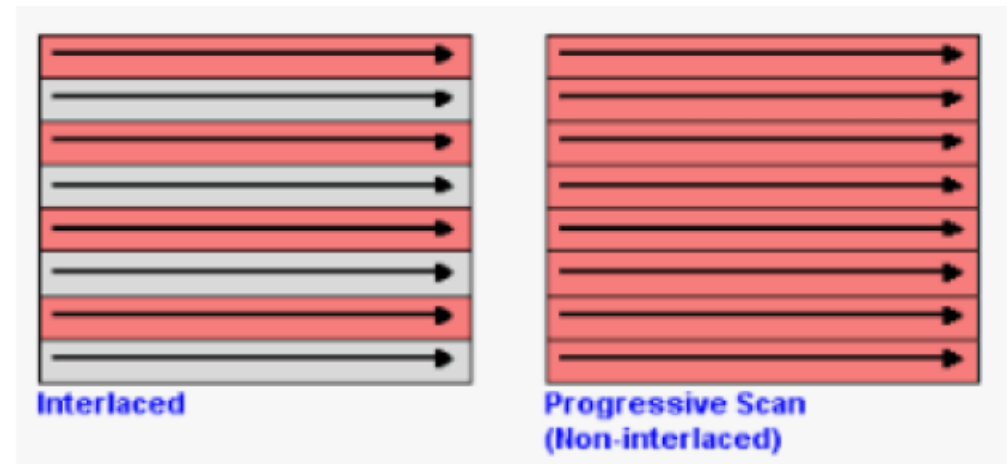
- Pixels become visible
- Edges look **blocky or jagged**
- This effect is called **pixelation**

Smooth image - Before zoom  
Blocky image - After zoom

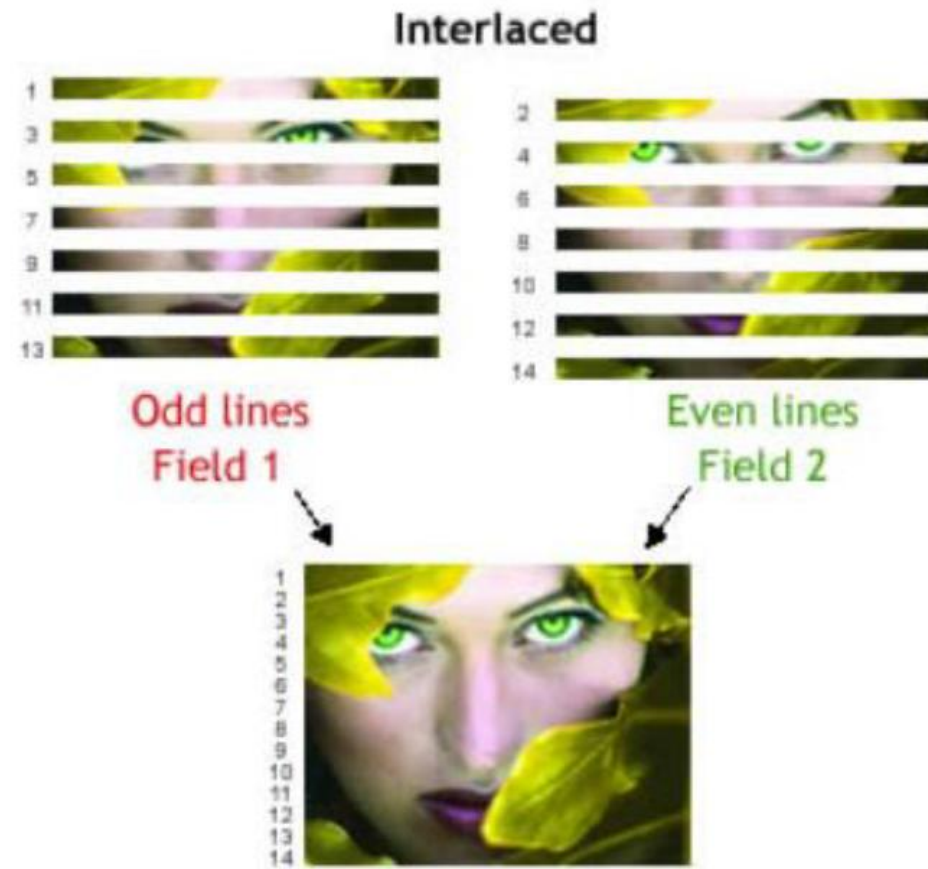


# Interlacing

- Interlacing is a way of displaying images on a screen. The screen does not draw all lines at once.
- First scan -Draws every other line (odd lines)
- Second scan - Draws the remaining lines (even lines)
- Together, these scans form the full image.
- Why Use Interlacing?
  - Uses less data per scan
  - Allows faster refresh rate
  - Cheaper to implement (used in older TVs)
- Disadvantages
  - Can cause flickering
  - May show visible moving lines



# Interlacing



Field 1 + Field 2 = Frame (complete image)

Display Rate: 60 fields per second (North America)

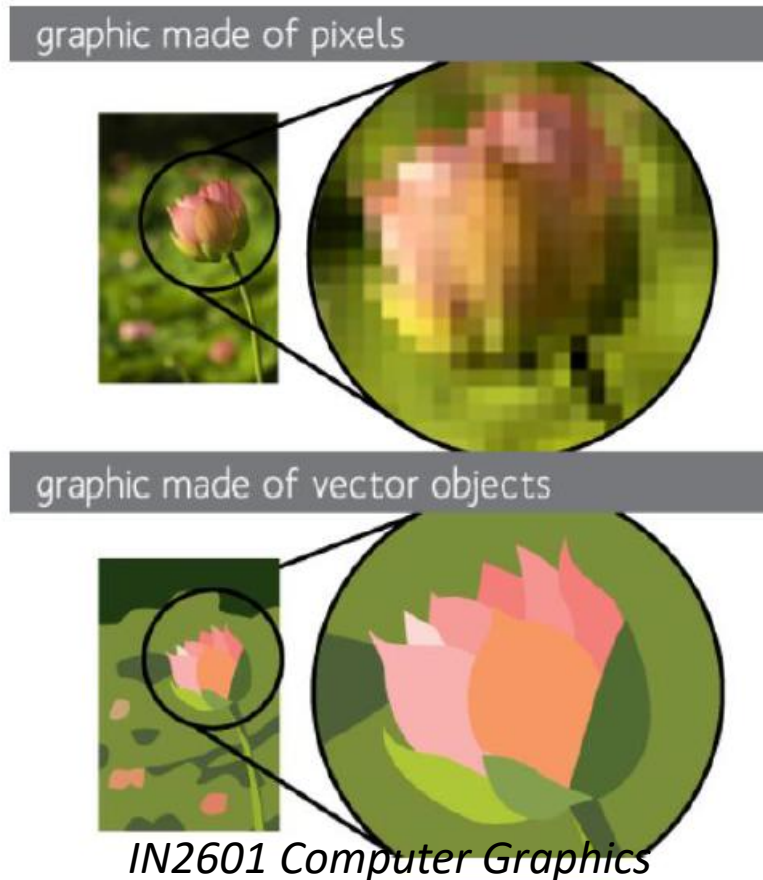
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# Vector Graphics

- Images created using **mathematical shapes** like:
  - Lines
  - Points
  - Curves
  - Polygons
- These shapes are defined using **mathematical equations**.
- Can be **resized or zoomed** without losing quality.
- Image stays smooth and sharp at any size.
- File size usually remains small.

# Vector Graphics

- Common Vector File Types .svg, .ai, .cdr, .eps



# Activity

**Compare and Contrast raster and vector graphics.**

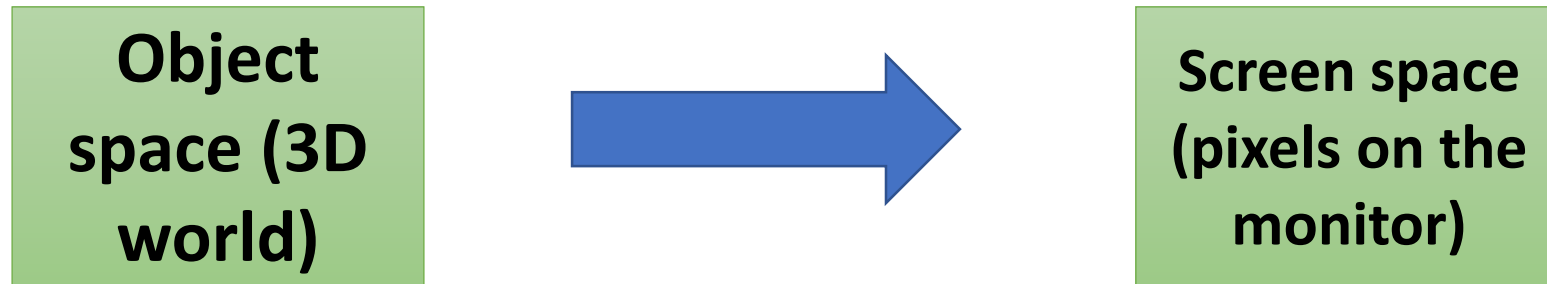


# Traditional Graphics Pipeline

- When a shape (primitive) is drawn on the screen, it goes through several steps,
  1. Transformation
  2. Lighting
  3. Clipping
  4. Scan conversion
  5. Pixel processing

# Transformation

- The transformation process refers to the linear transformation from 3D space to a 2D viewing space which simply means changing a 3D object so it can be shown on a **2D screen**.
- Every point (vertex) of the object must move from,



- **Types of Transformations**
  - **Translation** -Move the object
  - **Rotation** -Turn the object
  - **Scaling** -Change the size
  - **Perspective Transformation** -Make far objects look smaller (realistic view)

# Clipping

- **What is Clipping?**
- Some shapes are:
  - Fully visible on the screen
  - Completely outside the screen
  - Partly inside and partly outside
- If a shape is partly outside the screen, we **cut (clip)** the invisible part.

# Clipping

## View Volume

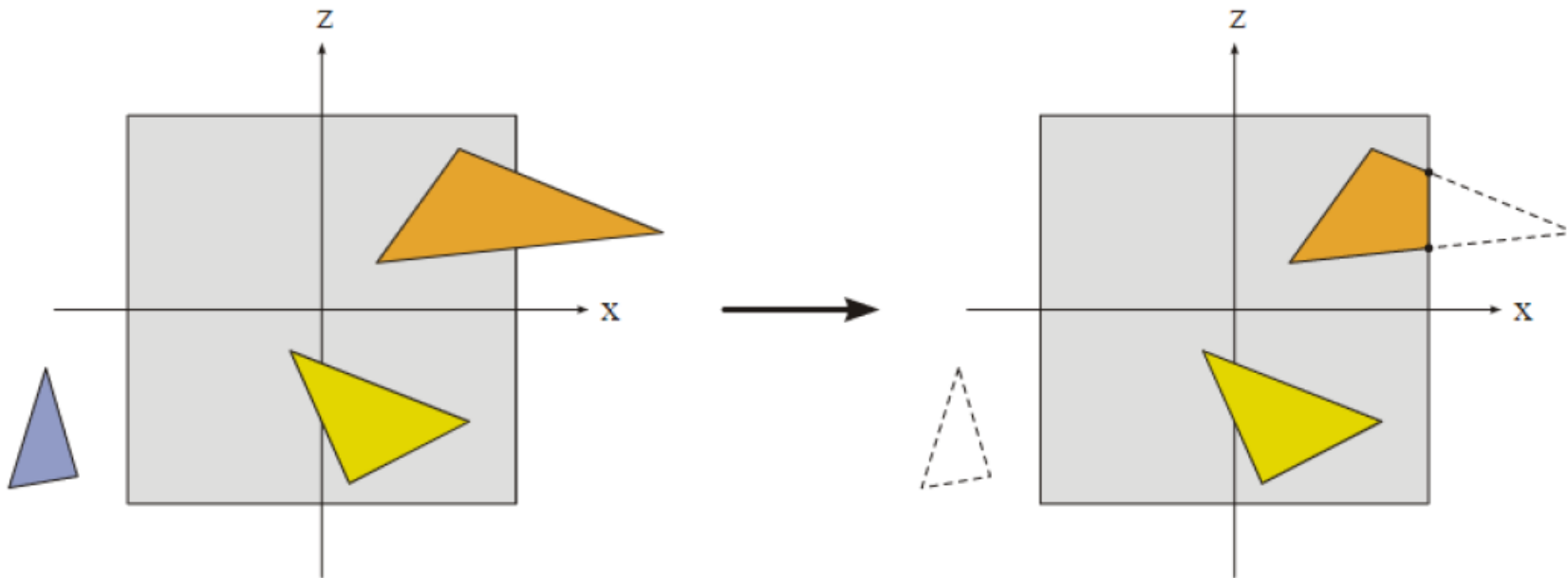
- The camera can only see a certain 3D area.
- This visible area is called the **view volume**.
- Objects outside this area are not shown.

## Culling

- **Culling** removes shapes that are completely invisible.
- Saves processing time.

# Clipping

The output of the clipping/culling process is a set of visible triangles that lie within the dimensions of the display device.



# Lighting

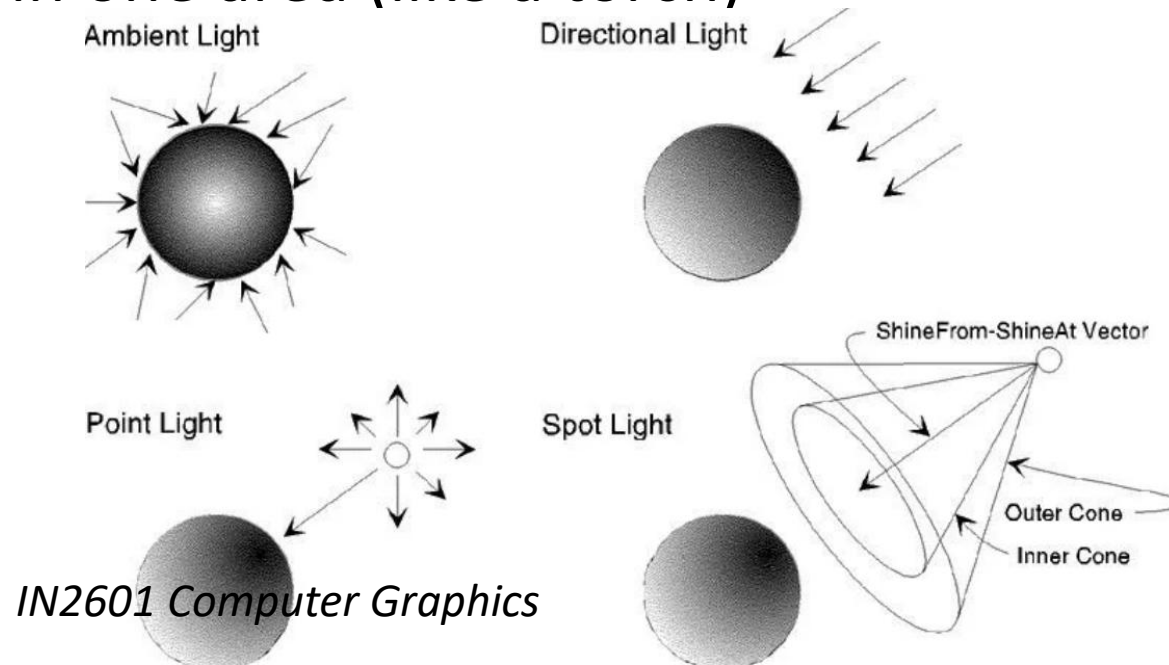
- Lighting decides the **color and brightness** of an object.
- Basic method:
  - Lighting is calculated at each **vertex (corner point)**.
- Advanced method:
  - Lighting is calculated for **each pixel** (more realistic).



# Lighting

## Types of Lights

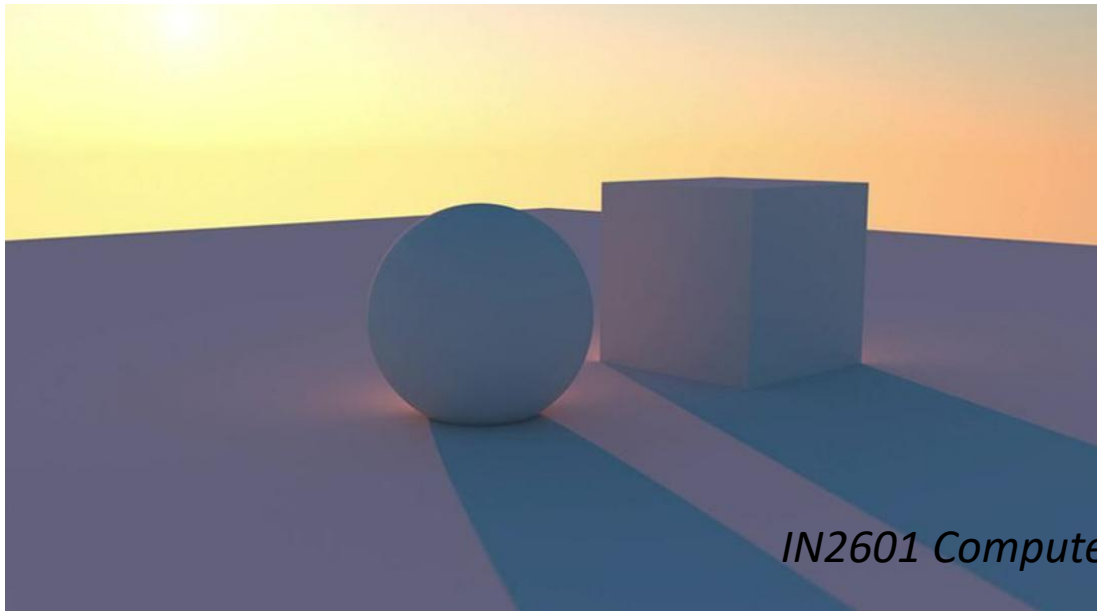
- **Point Light** - Light from one position (like a bulb)
- **Directional Light** - Light from one direction (like sunlight)
- **Spotlight** - Light focused in one area (like a torch)



# Lighting

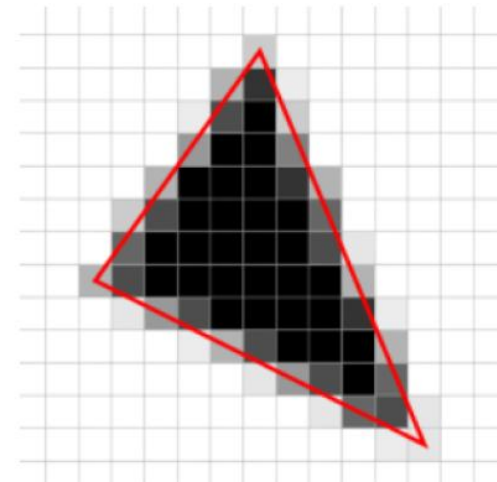
## Advanced Lighting Effects

- Shadows
- Reflections
- Transparency
- Realistic light bouncing



# Scan Conversion

- **Scan conversion** is the process of converting geometric primitives (such as lines or triangles) defined in continuous coordinate space into a discrete set of pixels on a raster display.
- Each corner (vertex) of the shape has a color.
- The colors are smoothly **blended (interpolated)** across the shape.
- This means each pixel inside the shape can have a slightly different color (each pixel may have a unique color)



# Pixel Processing

- After scan conversion, we get many **pixels (x, y positions)**.
- Each pixel also has extra information like:
  - Color
  - Depth (z-value)
  - Texture data
- Pixel processing decides the **final color** of each pixel before showing it on the screen.
- **Z-Buffer (Depth Test) Technique** - Each pixel stores a **depth value (z-value)**.
  - Used to decide **which object is in front**.
  - A pixel is drawn only if it is **closer to the camera**.
  - Prevents hidden objects from appearing.

# Exercise

- A triangle is partly outside the screen and overlaps another triangle behind it.
- Which stages handle,
  1. Cutting the outside part?
  2. Deciding which triangle appears in front?
  3. Converting it to pixels?

# Exercise

- Why can't we directly display a 3D object on the screen?

# Exercise

- Which stage is most important for realism? Why?



# Questions?